Global CO_2 Emissions & Temperature Change

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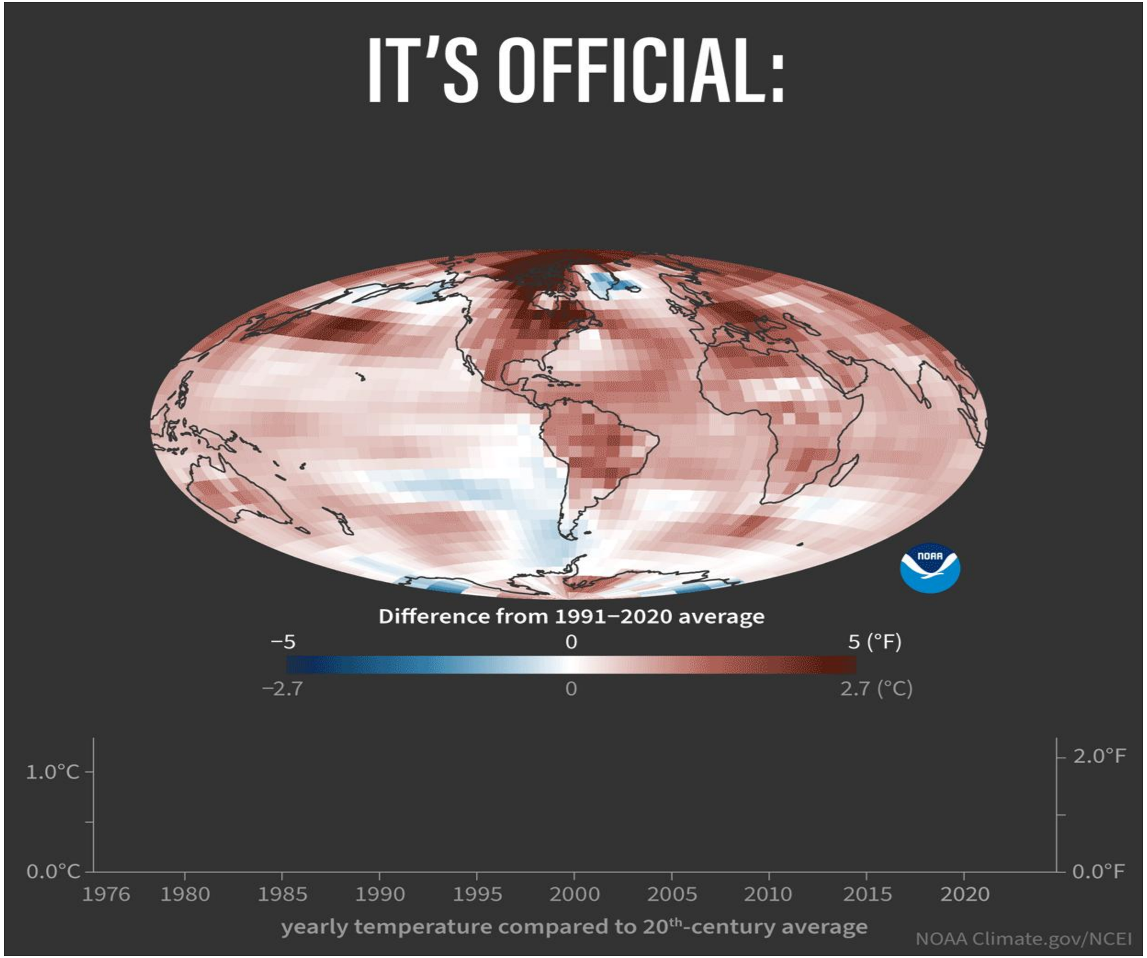
Global warming
a major public concern

Global temperature is an indicator of changes in Earth's energy balance:

it controls:

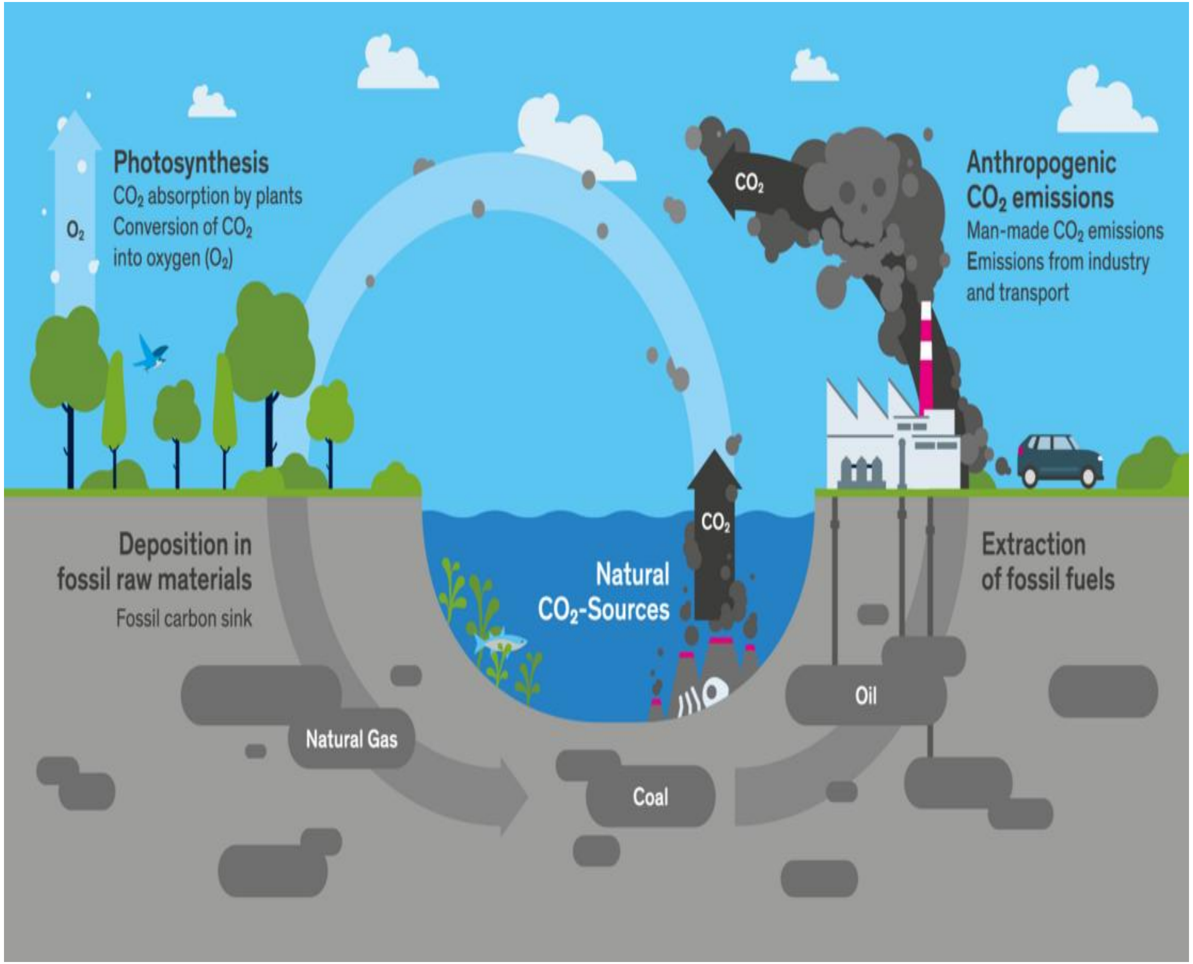
- Water cycle (evaporation, clouds, surface water supplies, and precipitation)
- Carbon cycle
- Plants, animals, and other living things that can survive in different places on Earth



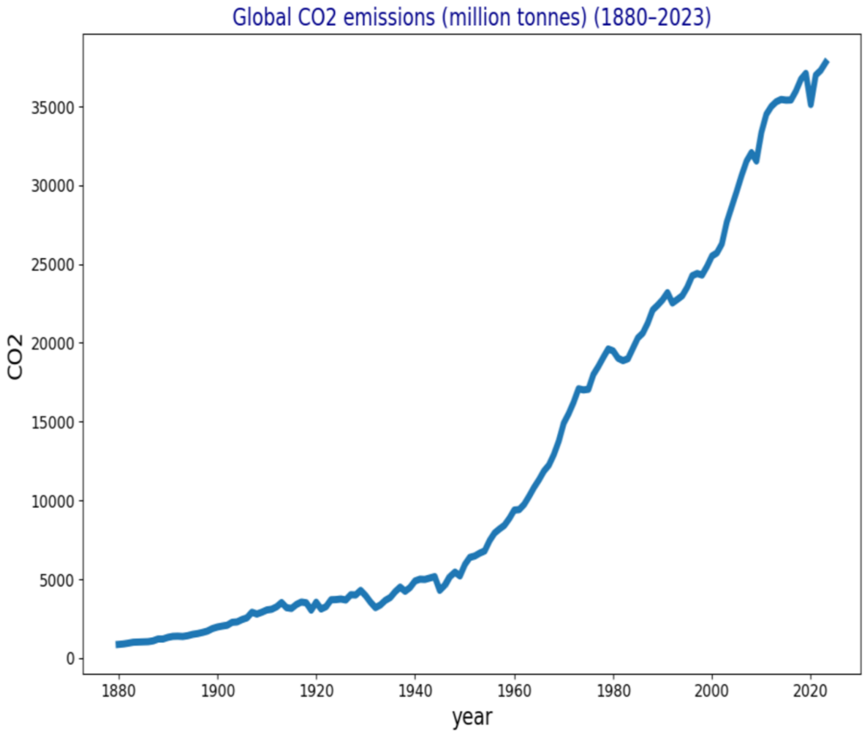
CO_2 Emissions
A key driver of climate change

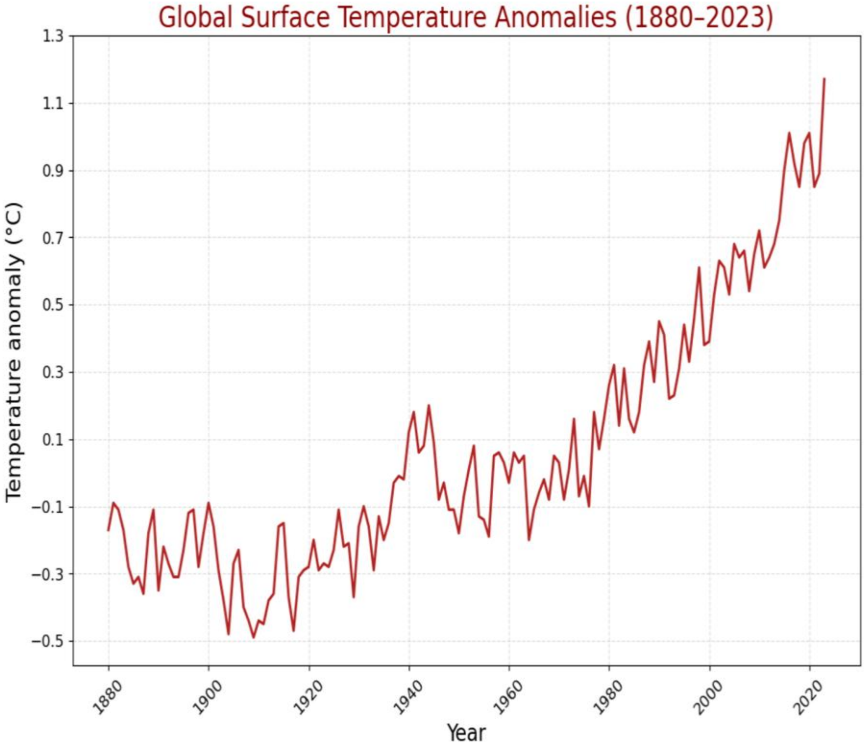
Carbon dioxide (CO_2) is a greenhouse gas that traps heat in Earth's atmosphere. It comes mainly from:

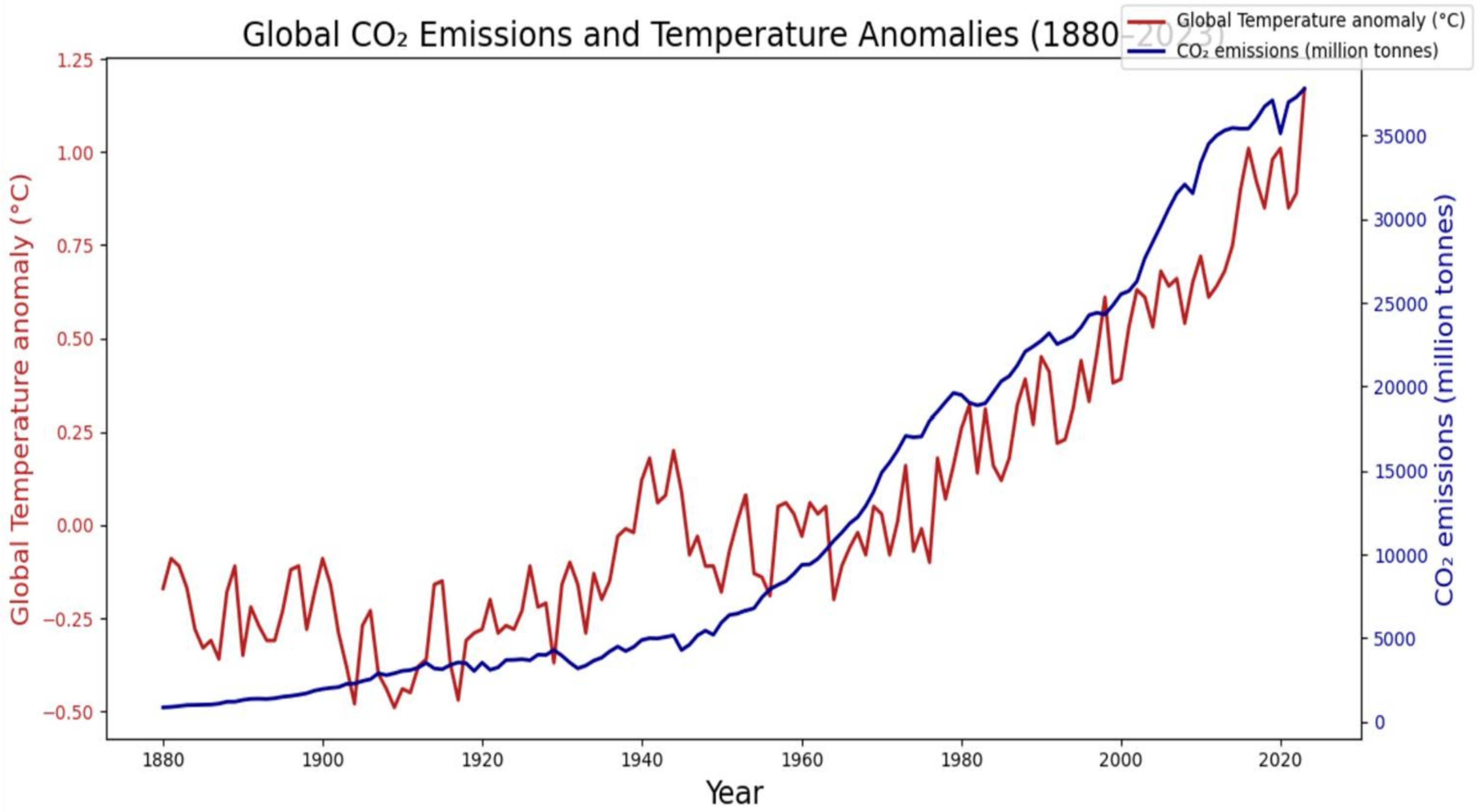
- Burning fossil fuels (coal, oil, and natural gas)
- Industrial activities and deforestation



Earth Surface Temperature Trend & CO2

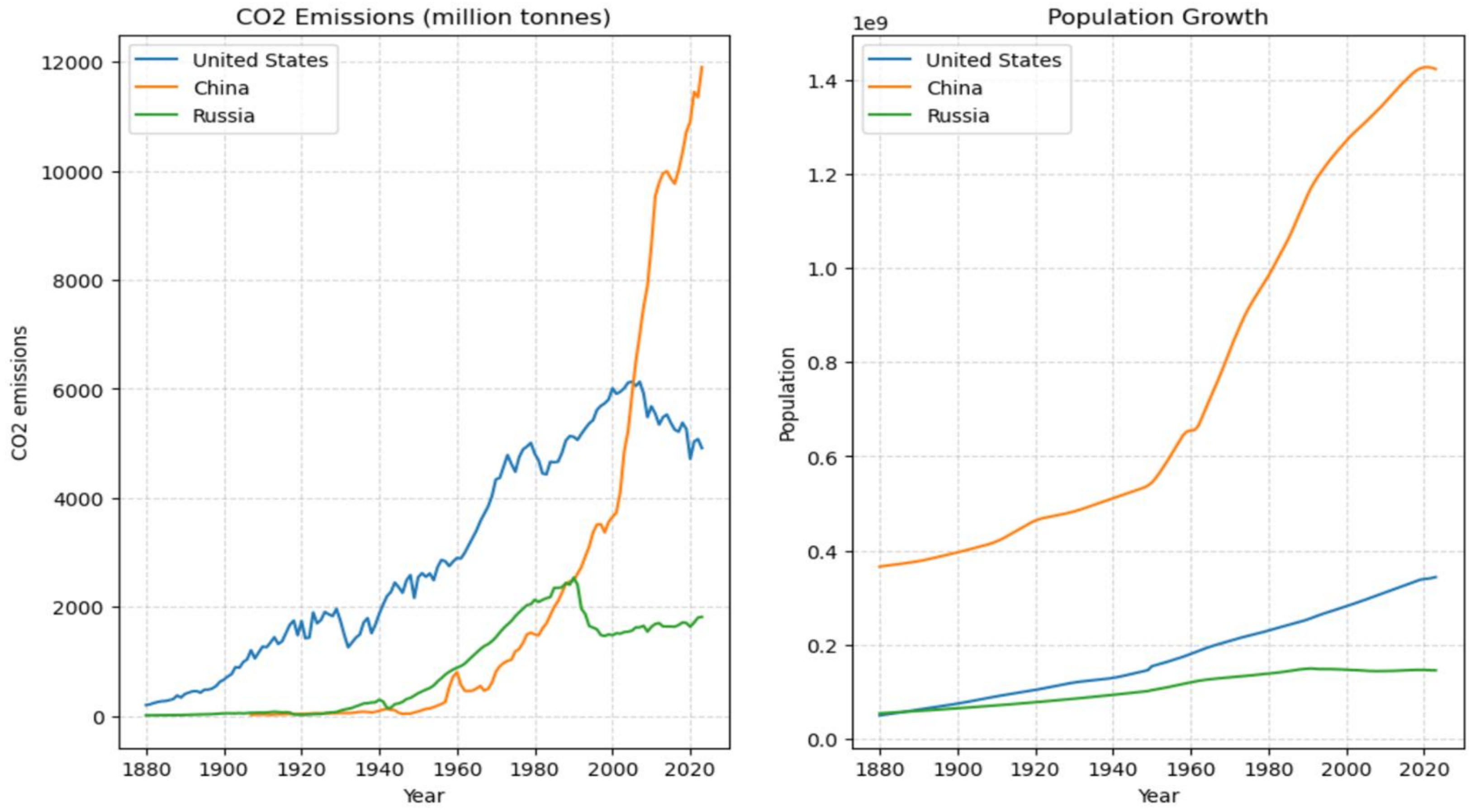




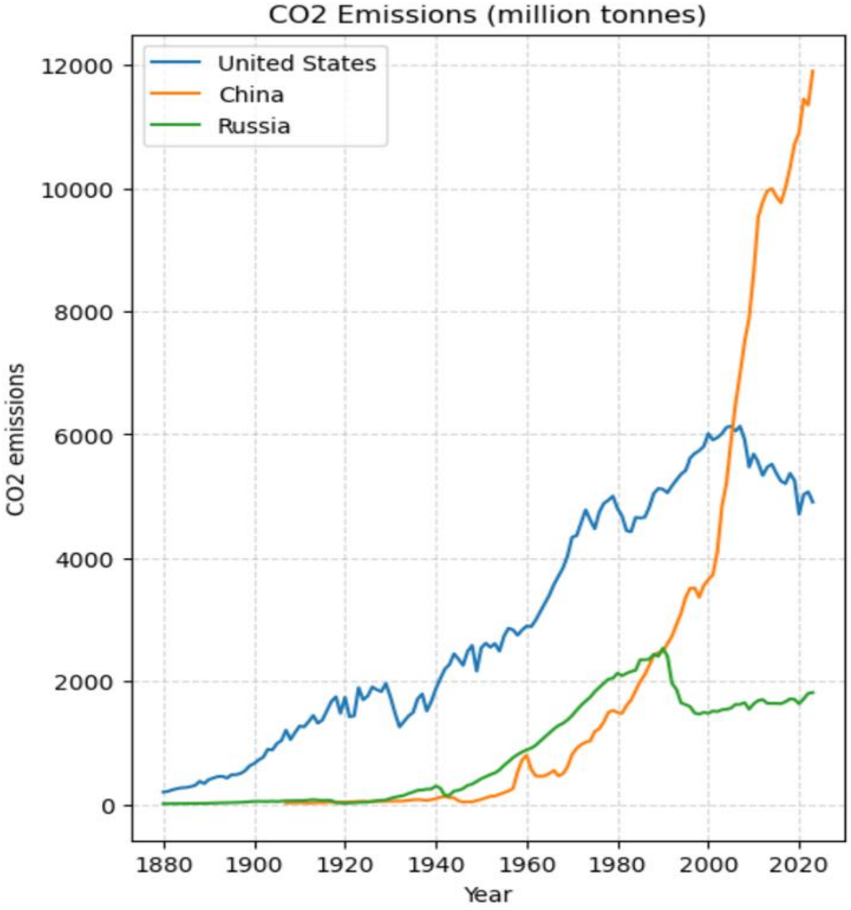


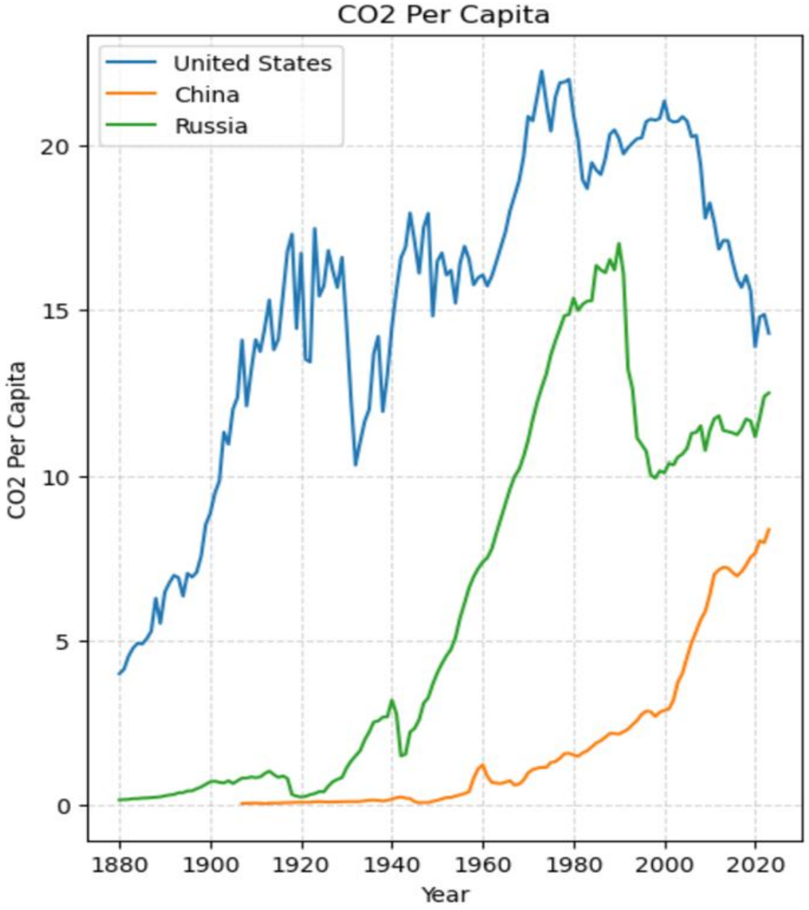
Global CO₂ Emissions and Temperature Anomalies (1880–2023)
Global Temperature anomaly (°C)
CO₂ emissions (million tonnes)
Global Temperature anomaly (°C)
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
CO₂ emissions (million tonnes)
35000
30000
25000
20000
15000
10000
5000
0
1880
1900
1920
1940
1960
1980
2000
2020
Year

Total CO_2 emissions by the top 3 emitting countries & their population growth rates



Total and Per-Capita CO_2 Emissions by the top 3 emitting countries





Annual CO_2 Growth Rate

How Fast CO_2 Has Been Growing Each Year

- The average annual CO_2 growth rate is about **2.9%**.
- The highest was **16.6% in 1920** (industrial recovery after World War I)
- The lowest **–16.9% in 1945** (World War II impact).

Annual growth of CO2

```
world = co2[co2['country'] == 'World'][['Year', 'co2_growth_prct']].dropna()

avg_growth= world['co2_growth_prct'].mean()
max_growth= world['co2_growth_prct'].max()
min_growth= world['co2_growth_prct'].min()

year_max=world[world['co2_growth_prct']==max_growth]["Year"].iloc[0]
year_min=world[world['co2_growth_prct']==min_growth]['Year'].iloc[0]

print(f"Average annual CO2 growth rate:{avg_growth:.2f}%")
print(f"Highest growth:{max_growth:.2f}% in {year_max}")
print(f"Lowest growth:{min_growth:.2f}% in {year_min}")

#This shows the average annual CO2 growth rate is 2.87 percent which shows CO2 emissions have increased moderately every year on average since 1880.
#Highest growth is 16.61% in 1920 and the lowest growth is -16.91% in 1945.
#I searched these dates and figured out maybe the world wars was influential on these percentages.
#world war 1 was from (July 28, 1914, to November 11, 1918) and in 1920 the highest growth shows industrial recovery and world war 2 was from
#(September 1, 1939, to September 2, 1945) which the lowest percentage of growth was for 1945, the last year of world war!
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Average annual CO2 growth rate:2.87%
Highest growth:16.61% in 1920
Lowest growth:-16.91% in 1945
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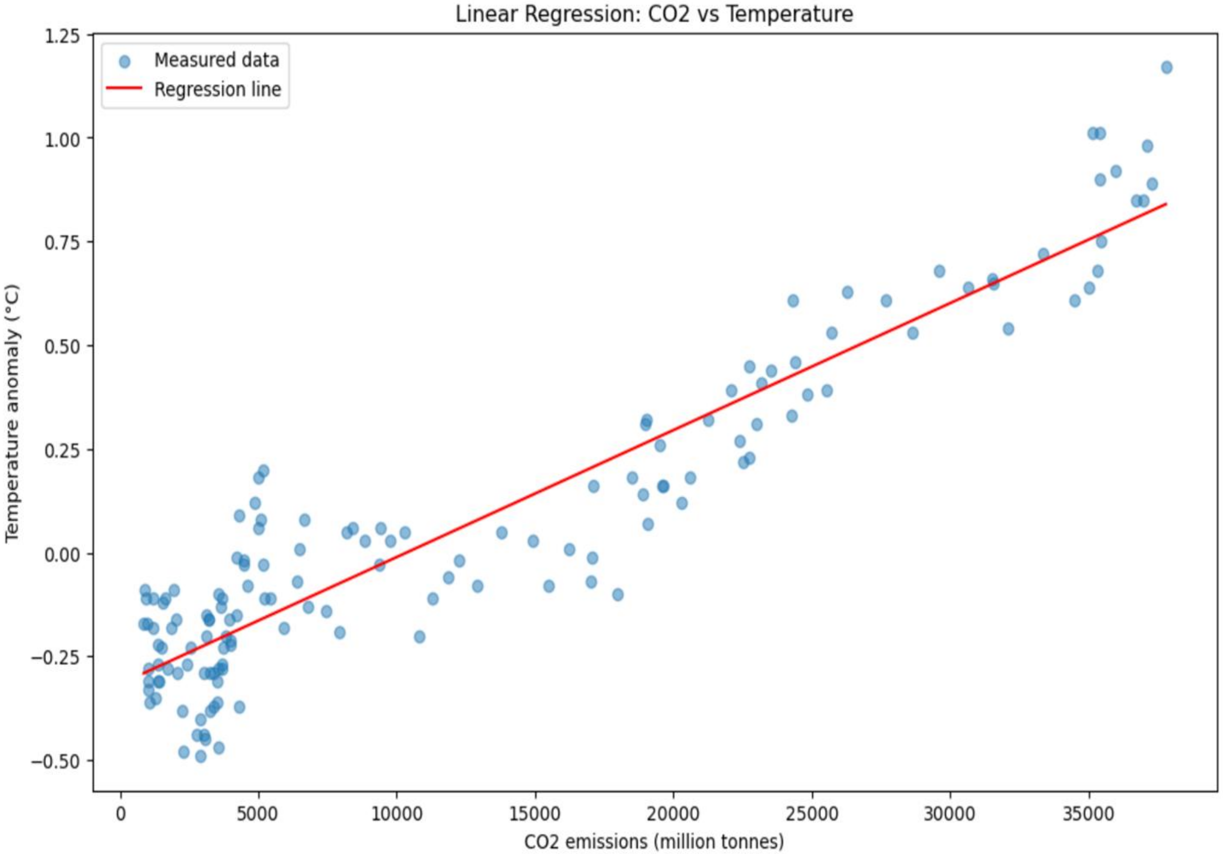
Linear Relationship Between CO_2 Emissions and Global Temperature



The **blue** points show real data from 1880–2023, and the **red** line is the regression model fitted to these points.
It shows a **clear positive linear relationship**:

- When **CO_2** emissions increase, the Earth's **temperature** also rises.
- The model line is close to the data points → **CO_2 explains most of the warming trend**.
- This **confirms** that human activities releasing **CO_2** strongly affect **global climate**.

Conclusion

- This analysis showed that there is a clear relationship between CO_2 emissions and global temperatures.
- From 1880 until 1950, both CO_2 emissions and temperatures were increasing slowly.
- After 1950 they began to increase much more rapidly.
- China now has the highest CO_2 emissions overall, however the United States still has the highest per capita emissions.
- This means that population alone is not the cause of global warming, so energy consumption and industry also have big impact.
- The average annual growth rate of CO_2 is about 2.9%, this shows that CO_2 emissions have increased steadily almost every year since 1880.
- The fastest growth was in 1920 with 16.6%, when industries expanded after World War I.
- The slowest was in 1945 with -16.9%, at the end of World War II.
- I also used a linear regression model to further convince myself that global temperature increases are strongly related to CO_2 emissions.
- This model also shows statistically that increases in CO_2 play a major role in global warming.

Thank you